



School of Engineering and technology

Department of Electronics & Communication Engineering

Arduino Project

ON

“SMART HELMET SAFETY SYSTEM With Accident and Alcohol Detection”

Team Members

Prajwal Gowda C [24BBTCA086]

Prajwal Gowda J [24BBTCA087]

Surya Prathap [24BBTCA092]

Sanjay Kumar [24BBTCA106]

Project Guide

Prof. Silpa Sivan P

Assistant Professor

Dept. of ECE,

SoET, CMRU, Bangalore.

Date: 21/ 04 / 2026

Outline

- **Abstract**
- **Introduction**
- **Objective/Objectives**
- **Methodology**
- **Hardware Description**
- **Software Used**
- **Advantages**
- **Application**
- **Conclusion**
- **References**

Abstract

- This project focuses on improving road safety using a smart helmet system.
- It checks whether the rider is wearing a helmet and also detects alcohol consumption.
- If unsafe conditions are found, the system gives alerts using buzzer and led, and stops the vehicle using a relay.
- It also includes accident detection and can send alerts using gsm module. This system helps in reducing accidents and improving safety.

Introduction

- Road accidents are increasing day by day due to careless driving.
- Many riders do not wear helmets or drive after drinking alcohol. This project is designed to solve these problems using sensors and a microcontroller.
- The system ensures that the rider follows safety rules before starting the vehicle.

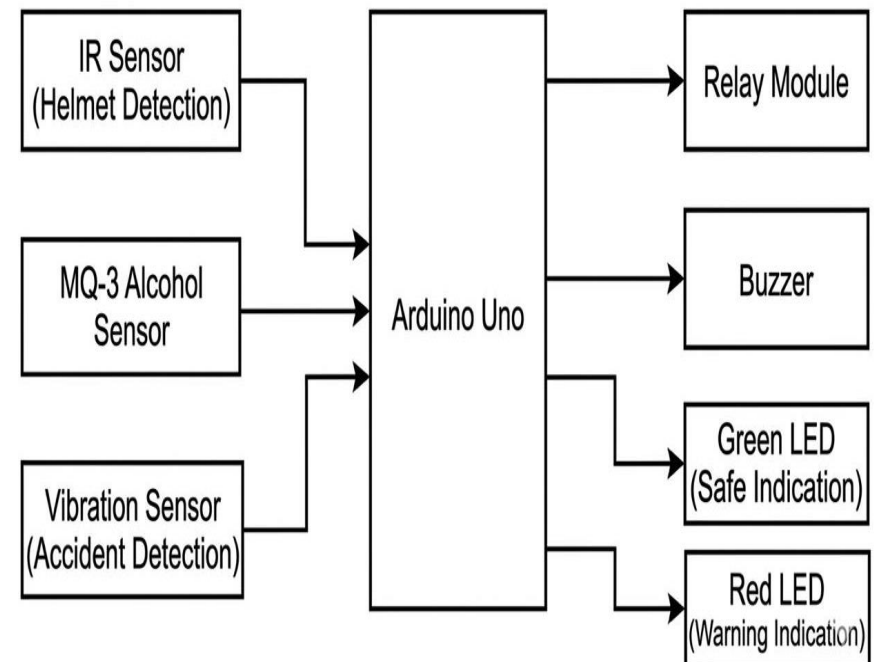


Objectives

- To detect whether the helmet is worn
- To detect alcohol consumption
- To prevent vehicle start in unsafe conditions
- To detect accidents using vibration sensor
- To send alert messages using GSM module

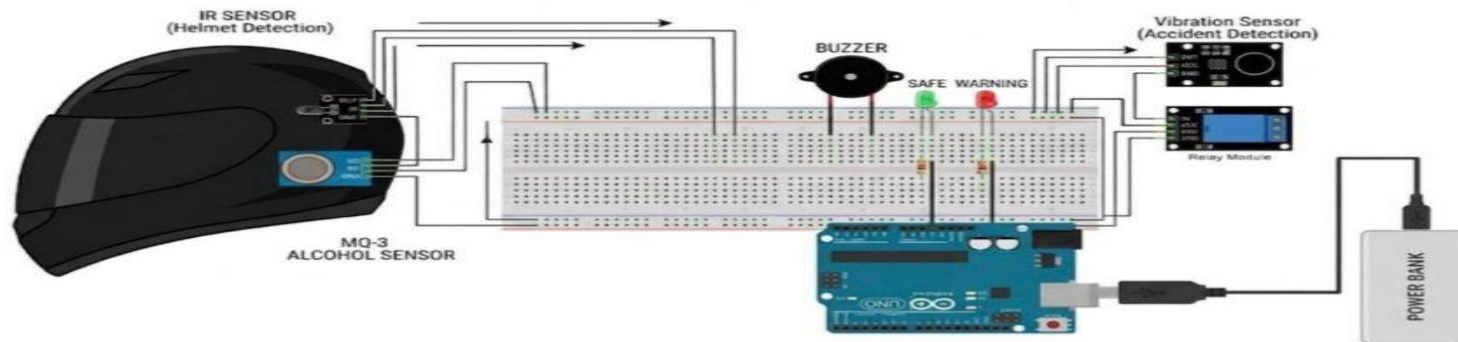
Methodology

- First, all sensors are connected to the Arduino. The IR sensor checks helmet wearing, and the MQ-3 sensor checks alcohol.
- The vibration sensor detects accidents. Based on these inputs, Arduino processes the data and controls outputs like LED, buzzer, and relay.
- If conditions are safe, the system allows operation; otherwise, it gives alert.



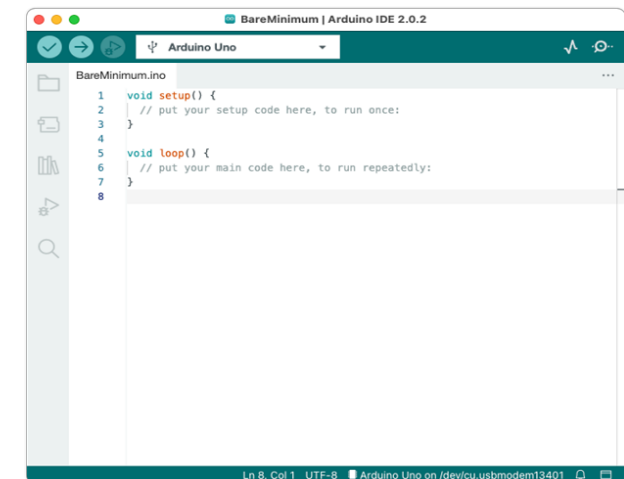
Hardware Description

- The main components used are Arduino Uno, IR sensor, MQ-3 alcohol sensor, vibration sensor, relay module, buzzer, LEDs, GSM module, breadboard, and connecting wires.
- Each component plays an important role in detecting and controlling the system.



Software Used

- The project is developed using Arduino IDE. The code is written in Embedded C language. The software is used to upload the program into the Arduino and control the working of sensors and output devices.



Advantages

- Improves rider safety
- Prevents drunk driving
- Ensures helmet usage Easy to use and cost-effective
- Can be extended with GPS and GSM

Application

- Used in two-wheeler safety systems
- Helps in reducing road accidents Useful for traffic monitoring
- Can be used in smart transportation systems

Conclusion

- This project successfully improves road safety by combining different sensors and technologies.
- It ensures that the rider follows safety rules and helps in preventing accidents.
- It is a simple and effective system that can be further improved using advanced technologies.

References

- [1] Arduino Official Website**
- [2] MQ-3 Sensor Datasheet**
- [3] IR Sensor Working Principle**
- [4] Embedded Systems Concepts**

THANK YOU